

Notes: Detragiache, Garella, and Guiso (JF, 2000)

Multiple versus Single Banking Relationships: Theory and Evidence

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1 Big picture

- **Research question.** Why do firms borrow from more than one bank? More specifically, when is a single relationship optimal, when is multiple banking optimal, and what determines the number of bank relationships?
- **Main idea.** A relationship bank learns private information about a firm's project. If that bank later suffers a liquidity problem and cannot roll over credit, the firm may have to refinance from uninformed banks. Those outside banks face adverse selection and may refuse to lend. Multiple relationships reduce the chance that all informed banks are simultaneously unable to refinance the project.
- **Core friction.** Relationship lending creates information advantages, but it also creates exposure to relationship-bank liquidity risk. Multiple banking is costly, yet it insures against premature liquidation when nonrelationship finance is unavailable.
- **Main theoretical result.** If uninformed banks are willing to refinance at the continuation date, the firm uses one bank. If uninformed banks refuse to refinance, the firm may optimally borrow from several banks to raise the probability that at least one informed bank remains active.
- **Empirical object.** The paper studies a cross-section of Italian small and medium-sized manufacturing firms, matched to credit-register data and bank balance-sheet data, to explain both the single-versus-multiple choice and the number of relationships conditional on multiple banking.
- **Main empirical finding.** The data are broadly consistent with the model: stronger loan enforcement and higher profitability make single banking more likely, while bank fragility raises the number of relationships in the multiple-banking region.
- **Economic takeaway.** Multiple banking is not only about limiting relationship-bank hold-up. It can also be a form of insurance against the possibility that an informed relationship bank cannot continue funding a profitable project.

2 Incremental contribution of the paper

- **Moves beyond the one-bank benchmark.** Much of the theory of relationship lending assumes a single relationship bank. The paper begins from the empirical fact that many firms, including small firms, use several banks.
- **Explains why more than two banks may arise.** Earlier theories can explain why a second bank may discipline a relationship lender or complicate renegotiation. This model gives a rationale for a larger number of relationships: more informed banks lower the probability that all informed lenders are inactive at the refinancing date.
- **Introduces liquidity-risk insurance.** The paper connects the structure of bank-firm relationships to Diamond-style liquidity risk: a profitable project may be liquidated early because informed banks face internal liquidity problems.

- **Combines theory and matched data.** The empirical work links firm characteristics, bank characteristics, relationship counts, and regional enforcement conditions using Italian survey, credit-register, and bank-balance-sheet data.
- **Separates two margins.** The empirical design distinguishes the decision to use a single versus multiple banking regime from the number of relationships chosen within the multiple-banking regime.
- **Provides cross-country motivation.** The paper documents that small firms in Italy have many more bank relationships than small firms in the United States, then uses Italian data because relationship counts vary widely across firms.

3 Theoretical model

3.1 Project technology and information

The firm has no internal funds and needs external finance for a two-stage project. At date 0 the project requires investment I_0 . At date 1 it requires a second investment I_1 to continue. At date 2 it pays K with probability p and zero otherwise.

- The project is worth continuing whenever even the lowest type has positive continuation value: $p_1 K - I_1 \geq 0$.
- Banks that finance the first stage learn the realized project quality p at date 1.
- Nonrelationship banks do not observe p at date 1.
- Loan enforcement is imperfect: creditors can extract at most a fraction $v < 1$ of project cash flow.
- Each banking relationship requires a fixed processing cost c , so multiple banking is costly.

3.2 Bank liquidity risk

Each relationship bank may suffer a liquidity crisis at the refinancing date. Let ε denote the probability that a bank is unable to refinance the project. Liquidity shocks are independent across banks.

- If the firm has n relationship banks, the probability that all are inactive is ε^n .
- The probability that at least one informed bank is active is therefore $1 - \varepsilon^n$.
- Increasing n lowers liquidation risk, but raises transaction costs by adding more relationships.

3.3 Refinancing by informed and uninformed banks

At date 1 the firm must repay the first-period loan and finance I_1 . If an informed bank is active, it knows p and can refinance the project when expected repayment is sufficient. The

maximum enforceable interest factor is:

$$R^* = \frac{vK}{D + I_1},$$

where D is the repayment due on the first-period loan.

An uninformed bank faces an adverse-selection problem. A firm seeking outside refinancing may be good but unlucky because all relationship banks are in crisis, or it may be bad and rejected by informed lenders. The outside bank refinances only if its expected profit is nonnegative at the maximum enforceable rate:

$$R^*(D + I_1)E^u(p \mid R^*) - (D + I_1) > 0.$$

Interpretation. This condition is the dividing line between single and multiple banking. If uninformed finance is feasible, the firm does not need multiple informed lenders. If uninformed finance is unavailable, additional relationship banks become valuable insurance.

3.4 Single-banking equilibrium

If uninformed banks are willing to refinance at date 1, the first-period loan is safe and the firm can always continue the project. Expected profit simplifies to:

$$\pi(n) = E(p)K - (I_0 + I_1 + nc).$$

Because n only adds transaction costs in this region, the firm chooses one relationship.

- **Proposition 1.** If nonrelationship refinancing is possible, single banking is optimal.
- **Economic intuition.** Once outside refinancing is available, multiple relationships provide no insurance benefit and only duplicate fixed costs.

3.5 Multiple-banking equilibrium

If uninformed banks refuse to refinance, the project can continue only if at least one informed bank is active and the project is sufficiently good. Expected profit in the multiple-banking region is:

$$\pi(n) = (1 - \varepsilon^n)Z - (I_0 + cn),$$

where

$$Z = \int_{I_1/vK}^1 (pK - I_1)f(p) dp.$$

Maximizing over n yields:

$$n^* = \frac{\ln c - \ln(-Z \ln \varepsilon)}{\ln \varepsilon}.$$

- **Benefit of more banks:** lower probability of premature liquidation, because $1 - \varepsilon^n$ increases with n .
- **Cost of more banks:** higher relationship setup costs, cn .
- **Key trade-off:** multiple banking is valuable only when relationship-bank liquidity risk and adverse selection in outside refinancing are important enough.

3.6 Comparative statics

- **Loan enforcement.** More efficient enforcement, higher v , makes outside lenders more willing to refinance, so it expands the single-banking region. Conditional on being in the multiple-banking region, it can increase the optimal number of relationships because continuation becomes more valuable.
- **Bank fragility.** Higher ε can make single banking more likely because outsiders become less suspicious that a firm seeking refinancing is a lemon; the firm may simply be unlucky because relationship banks are in crisis. Conditional on multiple banking, higher bank fragility tends to increase the desired number of relationships.
- **Project profitability.** More profitable projects make refinancing easier and make the cost of liquidation larger. Profitability therefore affects both the regime choice and the number of relationships.
- **Nonmonotonicity.** The number of relationships can be one for very low bank fragility, rise in an intermediate multiple-banking region, and return to one when bank fragility is high enough that outside refinancing becomes feasible.

4 Data and measurement

4.1 Italian firm-bank matched data

The empirical analysis combines three data sources.

- **Survey of Manufacturing Firms (SMF).** The 1994 wave of the Mediocredito Centrale survey covers small and medium-sized manufacturing firms and includes employment, ownership, location, R&D, innovation, sales, exports, investment, and balance-sheet information.
- **Credit Register (CR).** The Bank of Italy credit-register data identify each bank lending to each firm and the loan categories used. A firm is treated as having a relationship with a bank if it borrows from that bank in at least one loan category.
- **Bank Balance Sheet database (BB).** This provides bank-level balance-sheet and institutional characteristics, allowing the authors to construct bank fragility and bank size measures for each firm's lenders.

The merged reference sample contains 1,849 firms. The empirical regressions in Table IV use roughly 1,754 to 1,760 observations for the single-banking probit and roughly 1,557 to 1,563 observations for the multiple-banking second stage.

4.2 Dependent variables

- **Single-banking indicator.** A dummy equal to one if the firm borrows from only one bank.

- **Number of relationships.** The number of banks lending to the firm, estimated only for firms in the multiple-banking regime.

4.3 Main explanatory variables from the model

- **Loan-enforcement efficiency:** average share of defaulted loans recovered through legal procedures in the firm’s province. This proxies for v .
- **Bank fragility:** two measures are used. The first is the average variability of idiosyncratic liquidity shocks for the banks lending to the firm. The second is the average ratio of nonperforming loans to loanable funds among relationship banks.
- **Profitability:** gross profits divided by sales.

4.4 Additional controls

The paper also includes controls motivated by alternative theories and practical determinants of banking structure.

- Firm size, leverage, age, and group membership.
- Average size of relationship banks, interpreted partly as a proxy for bank liquidity.
- Entrepreneurial control-rent proxies: share owned by the largest owner and industry comovement.
- Innovation and R&D proxies: patents, R&D activity, product innovation, process innovation, and joint product/process innovation.
- Regional dummies for firm location.

5 Methodology and empirical specifications

5.1 Two-stage empirical design

The model implies two distinct empirical margins, so the paper uses a two-stage strategy.

1. **Stage 1: regime choice.** Estimate a probit model for whether a firm uses a single relationship:

$$\Pr(\text{Single}_i = 1) = \Phi(X_i\beta),$$

where X_i includes loan recovery, bank fragility, profitability, firm characteristics, bank characteristics, and regional controls.

2. **Stage 2: number of relationships.** For firms with multiple relationships, estimate the number of relationships using OLS with a selection correction:

$$N_i = X_i\gamma + \rho\lambda_i + u_i,$$

where λ_i is the Mills ratio computed from the first-stage probit.

5.2 Expected signs

- Higher loan recovery should increase the probability of single banking, but has weak or ambiguous implications for the number of relationships conditional on multiple banking.
- Higher bank fragility should make single banking more likely in the baseline monotonic prediction, but the model also predicts possible nonmonotonicity. Conditional on multiple banking, higher fragility should increase the number of relationships.
- Higher profitability should make single banking more likely and, conditional on multiple banking, can increase the desired number of relationships because the value of continuation is larger.
- Larger firms and more leveraged firms are expected to be less likely to rely on a single relationship and to have more relationships.
- Larger relationship banks should be associated with fewer relationships if bank size proxies for lower fragility and greater liquidity.

5.3 Nonlinear bank-fragility specification

Because the model predicts a nonmonotonic relationship between bank fragility and the single-banking probability, the paper also estimates an extension with truncated squared liquidity-shock terms. The idea is to let very low and higher levels of bank fragility have different effects on the probability of single banking.

6 Identification and interpretation

6.1 What the paper can identify

The paper provides evidence that observed banking-relationship structures line up with the model's comparative statics. The strongest evidence concerns cross-sectional associations between relationship counts and model-motivated proxies for enforcement, bank fragility, profitability, and firm characteristics.

6.2 Why the Italian setting is useful

Italy is useful because multiple banking is widespread and relationship counts vary substantially across firms. The credit register also allows the authors to observe all banks extending credit lines to each firm, which is essential for measuring the dependent variable and constructing lender-level fragility proxies.

6.3 Endogeneity concerns

The empirical design is not a randomized experiment. Several important variables may be jointly determined with relationship structure.

- Firms choose banks, so the average fragility of a firm’s relationship banks may partly reflect firm-bank matching.
- Regional loan-recovery efficiency may be correlated with other regional economic or institutional features.
- Profitability and leverage may be simultaneously related to both credit demand and bank willingness to lend.

The paper addresses these concerns mainly through controls, regional dummies, and a model-based two-stage structure rather than through a fully causal identification design.

6.4 Interpretation of the coefficients

The probit coefficients should be read as evidence about which firms are more likely to choose the single-banking regime. The OLS coefficients in the second stage should be read as evidence about how relationship counts vary among firms that use multiple banking, after correcting for selection into that regime.

6.5 Alternative explanations

The paper acknowledges and controls for several nonmodel mechanisms.

- Large firms may use multiple banks because they need larger loans or because banks diversify credit risk.
- Multiple banking may reduce informational monopoly rents of a relationship bank.
- Firms with more complex operations may naturally need more lenders.
- R&D-intensive firms may prefer fewer banks if multiple relationships raise information-leakage concerns.

7 Tables and figures

7.1 Table I and Figure 1

Read Table I:

- Small firms in the United States often use one or two bank relationships: the mode is one, the median is two, and 44.5 percent have one relationship.
- Italian small firms use many more relationships: the mode is three, the median is five, and only 11.0 percent have one relationship.
- This contrast motivates the paper’s central question: why do firms use multiple banks, especially when doing so is costly?

Read Figure 1:

- The figure plots the optimal number of relationships n as a function of bank liquidity-crisis probability ε .
- For very low ε , the insurance value of additional banks is small, so one relationship may be optimal.
- In the intermediate region, multiple banking becomes optimal and the number of relationships rises with bank fragility.
- When ε becomes high enough, outside lenders infer that a refinancing request is likely due to bank liquidity trouble rather than bad project quality, so the firm can again use a single relationship.

Economic message: Multiple banking is most valuable in the intermediate region where relationship-bank liquidity risk matters but uninformed outside finance is still unavailable.

Table I
Number of Bank Relations for Small Businesses
in Italy and in the United States

The data refer to firms with 500 or fewer employees. For the United States, they come from the 1988 wave of the National Survey of Small Business Finances. The number of relationships is the number of credit institutions (commercial banks, savings and loans associations, savings banks, credit unions, or mortgage banks) lending to the firm. For Italy, data are from the 1994 Survey of Manufacturing Firms and from the Credit Register. The number of relationships is the number of banks extending credit lines to the firm.

	United States	Italy
Mode	1	3
Median	2	5
75th percentile	2	8
Share of firms with one relationship	44.5%	11.0%

Table I: Number of bank relations for small businesses in Italy and the United States.

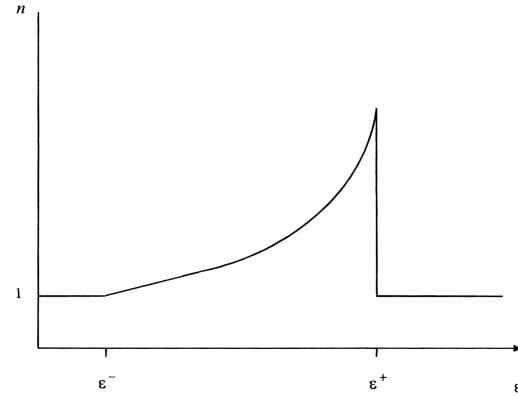


Figure 1. The optimal number of banking relationships. The figure shows the optimal number of relations (n) as a function of the value of ε , the probability of a liquidity crisis. ε^- and ε^+ are the thresholds for ε that define the multiple banking area. From $\varepsilon^- < \varepsilon < \varepsilon^+$ firms choose to borrow from multiple banks; for very low values of ε ($\varepsilon < \varepsilon^-$) or very high ($\varepsilon > \varepsilon^+$) a regime of single banking prevails.

Figure 1: Optimal number of banking relationships as bank liquidity risk changes.

7.2 Table II and Table III

Read Table II:

- Table II simulates the model under different profitability, enforcement, transaction-cost, and bank-fragility assumptions.
- “S” denotes the single-banking outcome.
- With low setup costs and fragile banks, the optimal number of relationships can be large, reaching 10 in the reported simulations.
- Higher transaction costs reduce the optimal number of relationships.
- Enforcement and profitability mainly shift the boundaries of the multiple-banking region; bank fragility affects both the region and the number of relationships.

Read Table III:

- Table III translates the theory into expected signs for the empirical regressions.
- Loan recovery, liquidity shocks, nonperforming loans, and profitability are expected to increase the probability of single banking.
- Conditional on multiple banking, bank fragility and profitability are expected to increase the number of relationships.
- Firm size and leverage are expected to reduce single banking and increase the number of relationships.

Economic message: The theory generates different predictions for the regime margin and the relationship-count margin. This is why the paper estimates the probit and the second-stage relationship-count regression separately.

Table II
Regions of Multiple Banking and Optimal Number of Relations for Various Parameter Values

The table reports the optimal number of relationships for various values of the parameters of the theoretical model. The single banking outcome is denoted by S . Also reported are the values of the parameter ϵ (denoted by ϵ^* and ϵ^{**}) marking the boundaries of the region in which the optimal number of relationships is greater than one. ϵ is the probability that a bank is hit by a liquidity shock. The parameter v is the maximum fraction of the cash flow that creditors can extract from the firm. c is the cost of setting up a relationship. I_0 and I_1 measure the investment costs of the project, and K is cash flow from the project if it succeeds. Given K , higher values of I_0 and/or I_1 imply a lower profitability of the project. All simulations assume that the probability of success of the project p is distributed uniformly in the interval $(0, 1)$.

Parameter Values		Boundaries of the Multiple Banking Region		Optimal Number of Relations for Different Values of the Probability of a Liquidity Shock (ϵ)		
		Lower Bound (ϵ^*)	Upper Bound (ϵ^{**})	$\epsilon = 0.2$	$\epsilon = 0.3$	$\epsilon = 0.5$
Panel A: Low Cost of Setting Up a Relationship $c = 1/1000$						
Low profitability ($K = 3, I_0 = I_1 = 0.2$)	$v = 0.5$	0.013	0.51	5	6	10
	$v = 0.6$	0.013	0.34	5	6	S
	$v = 0.8$	0.013	0.17	S	S	S
	$v = 0.9$	0.013	0.11	S	S	S
High profitability ($K = 3, I_0 = I_1 = 0.33$)	$v = 0.5$	0.016	1.00	5	6	10
	$v = 0.6$	0.016	1.00	5	6	10
	$v = 0.8$	0.018	0.54	5	6	10
	$v = 0.9$	0.018	0.43	5	6	S
Panel B: High Cost of Setting Up a Relationship $c = 1/100$						
Low profitability ($K = 3, I_0 = I_1 = 0.2$)	$v = 0.5$	0.013	0.53	3	4	6
	$v = 0.6$	0.013	0.36	3	4	S
	$v = 0.8$	0.013	0.18	S	S	S
	$v = 0.9$	0.013	0.13	S	S	S
High profitability ($K = 3, I_0 = I_1 = 0.33$)	$v = 0.5$	0.054	1.00	3	4	6
	$v = 0.6$	0.054	1.00	3	4	6
	$v = 0.8$	0.055	0.56	3	4	6
	$v = 0.9$	0.055	0.43	3	4	S

Table II: Regions of multiple banking and optimal relationship counts.

Table III
Explanatory Variables and Their Expected Sign

The table lists the explanatory variables used in the regressions and the expected signs of their coefficients according to the theoretical model. The notation (or zero) indicates that the variable has a negligible impact on the dependent variable according to simulation results. The share of defaulted loans recovered is the average share recovered through legal procedures by the banks operating in the firm's province. Liquidity shocks are defined as the average variability of the innovations to a measure of liquidity for the pool of banks lending to the firm. Nonperforming loans are scaled by loanable funds. Profitability is gross profits divided by sales. Leverage is the ratio of total bank debt to sales. Firm size is the number of employees. Firm age is the number of years since the firm was founded. For each firm the size of lending banks is the average value of loanable funds of its relationship banks. The comovement indicator is a measure of correlation of sales among firms in the same industry using a 44 industrial sector classification. See Appendix B for more detailed information on the construction of the variables.

Explanatory Variable	Probability of Single Banking	Number of Relationships in the Multiple Banking Region
Share of defaulted loan recovered	+	– (or zero)
Liquidity shocks	+	+
Nonperforming loans	+	+
Profitability	+	+
Firm size	–	+
Firm leverage	–	+
Firm age	–	+
Group membership	+	?
Average size of lending banks	–	–
Patents	–	+ (or zero)
R&D	–	+ (or zero)
Product innovation	–	+ (or zero)
Process innovation	–	+ (or zero)
Product and process innovation	–	+ (or zero)
Share of first owner	–	+ (or zero)
Industry comovement	–	+ (or zero)

Table III: Explanatory variables and expected signs.

7.3 Table IV

Read Table IV:

- Columns (1) and (2) estimate the probability of single banking using probit regressions.
- Columns (3) and (4) estimate the number of relationships in the multiple-banking region using OLS with a Mills-ratio selection correction.
- The share of defaulted loans recovered is positive and significant in the single-banking regressions, consistent with the prediction that better enforcement makes outside refinancing easier.

- Liquidity-shock measures are positive and significant in the second-stage relationship-count regressions, consistent with the prediction that firms exposed to more fragile banks use more relationships in the multiple-banking region.
- Profitability is positive and significant in the probit regressions, consistent with the prediction that profitable projects are easier to refinance outside the relationship.
- Firm size and leverage reduce the probability of single banking and increase the number of relationships.
- Average bank size is negatively related to single banking and, in one second-stage specification, negatively related to the number of relationships, supporting the interpretation that larger banks are less fragile or more liquid.
- The nonlinear liquidity-shock specification in column (2) supports the inverse-U pattern predicted by the model.

Economic message: The table is the main empirical test. It supports the paper's central mechanism: firms rely on multiple banks when relationship-bank liquidity risk and adverse selection make outside refinancing difficult.

Table IV
Determinants of the Choice between Single and Multiple Banking and of the Optimal Number of Banking Relationships in the Multiple Banking Region

The table reports regression coefficients and associated *t*-statistics. The first two regressions estimate the probability that the firm borrows from only one bank using a probit model. In the last two regressions the dependent variable is the number of relationships for firms using more than one bank. The regressions are estimated using OLS corrected for selectivity bias using the Mill's ratio computed from the first-stage probit. The Mill's ratios of regressions (3) and (4) are computed from the probit regressions (1) and (2) respectively. The share of defaulted loans recovered is the average share recovered through legal procedures by the banks operating in the firm's province. Liquidity shocks are measured by the average variability of the innovations to a measure of liquidity for the pool of banks lending to the firm. Nonperforming loans are scaled by loanable funds. Profitability is gross profits divided by sales. Leverage is the ratio of beginning-of-period total bank debt to sales. Firm size is the number of employees. Firm age is the number of years since the firm was founded. The comovement indicator is a measure of correlation of sales among firms in the same industry using a 44 industrial sector classification. See Appendix B for more detailed information on the construction of the variables. The (0,1) notation means that the variable is a dummy equal to one if the firm has the specified characteristic. All regressions include two regional dummies for the firm location.

Explanatory Variable	Probability of Single Banking		Optimal Number of Relationships in the Multiple Banking Region	
	(1)	(2)	(3)	(4)
Share of defaulted loans recovered	0.016 (2.175)	0.018 (2.454)	-0.001 (-0.068)	-0.015 (-0.971)
Liquidity shocks	0.504 (2.333)		2.141 (2.081)	2.510 (2.479)
(Liquidity shocks) ² <i>Q</i> > 10		-3.841 (-5.629)		
(Liquidity shocks) ² <i>Q</i> ≤ 10		1.657 (6.634)		
Nonperforming loans	5.589 (1.750)		26.016 (2.476)	9.759 (1.346)
(Nonperforming loans) ² <i>Q</i> ≤ 10		-17.465 (-0.822)		
(Nonperforming loans) ² <i>Q</i> > 10		15.558 (0.741)		
Profitability	2.164 (3.196)	1.757 (2.608)	0.834 (1.009)	0.053 (0.067)
Firm size	-0.187 (-3.221)	-0.135 (-2.241)	0.961 (7.276)	1.133 (9.581)
Firm leverage	-1.708 (-6.421)	-1.576 (-5.805)	3.968 (6.228)	4.739 (8.792)
Firm age	0.002 (0.902)	0.002 (0.831)	0.0215 (4.479)	0.0189 (4.026)
Group membership (0,1)	-0.017 (-0.163)	-0.019 (-0.175)	0.070 (0.325)	0.093 (0.436)
Average size of lending banks	-0.248 (-6.400)	-0.188 (-4.700)	-0.336 (-2.039)	-0.121 (-0.949)

Table IV—Continued

Explanatory Variable	Probability of Single Banking		Optimal Number of Relationships in the Multiple Banking Region	
	(1)	(2)	(3)	(4)
Patents (0,1)	-0.212 (-1.197)	-0.212 (-1.197)		
R&D (0,1)	-0.147 (-1.300)	-0.137 (-0.748)		
Product innovation (0,1)	-0.0429 (-0.278)	-0.0039 (-0.243)		
Process innovation (0,1)	-0.165 (-1.313)	-0.162 (-1.237)		
Product and process innovation (0,1)	0.117 (0.970)	0.109 (0.880)		
Share of first owner	-0.116 (-1.210)	0.002 (1.431)	0.005 (1.370)	0.003 (0.973)
Industry comovement	-0.163 (-0.910)	0.879 (1.368)	2.312 (1.712)	2.306 (1.737)
Constant	0.963 (1.371)	2.507 (2.759)	2.486 (1.125)	-0.640 (-0.351)
Mill's ratio			-10.446 (-7.289)	-7.506 (-10.01)
Number of observations	1,754	1,760	1,557	1,563

Table IV, part 2: Continued regression results.

Table IV, part 1: Determinants of single versus multiple banking and relationship counts.

8 Short summary

- The paper asks why firms use multiple banking relationships despite transaction and coordination costs.
- The central theory is that multiple banks insure the firm against relationship-bank liquidity crises.
- Relationship banks know project quality; outside banks do not, so outside refinancing can fail because of adverse selection.
- If outside refinancing is available, the firm uses one bank because extra relationships only add costs.
- If outside refinancing is unavailable, multiple relationships can reduce premature liquidation risk.
- The optimal number of relationships balances transaction costs against the probability that at least one informed bank remains active.
- The empirical analysis uses Italian matched bank-firm data from manufacturing-firm surveys, credit-register records, and bank balance sheets.
- The paper estimates a probit for single banking and an OLS selection-corrected regression for the number of relationships among multiple-bank firms.
- Better loan recovery and higher profitability make single banking more likely.
- Bank fragility increases the number of relationships among firms in the multiple-banking region.
- Firm size and leverage are strongly associated with more relationships, suggesting that practical credit-demand and risk-sharing forces also matter.
- The paper offers tentative cross-country evidence that less efficient loan recovery is associated with less single banking.

9 Conclusion

9.1 Core conclusion

Detragiache, Garella, and Guiso provide a theory and evidence for why firms, especially small and medium-sized firms, may borrow from multiple banks. Multiple banking can be optimal because it reduces the probability that a profitable project is liquidated when relationship banks face liquidity problems and outside lenders are unwilling to refinance because of adverse selection.

9.2 Economic intuition

A relationship bank's private information is valuable, but it also makes the firm vulnerable. If the relationship bank cannot continue lending, the firm must approach outsiders who do not know whether the firm is good-but-unlucky or bad. Multiple relationships create a pool of informed lenders, increasing the chance that at least one informed bank is available when refinancing is needed.

9.3 Empirical implication

The theory implies different predictions for the choice between single and multiple banking and for the number of relationships conditional on multiple banking. The empirical design follows this logic by estimating these two margins separately. The results are generally consistent with the model, especially for the role of loan recovery, bank fragility, and profitability.

9.4 Limitations

The evidence is cross-sectional and observational. Firm-bank matching, regional institutional differences, and unobserved firm complexity could also shape the number of relationships. The selection correction and rich controls help align the empirical work with the model, but they do not fully eliminate endogeneity concerns.

9.5 Takeaway

The enduring takeaway is that the number of banking relationships is part of the firm's risk-management problem. Multiple banking can be costly, but it can protect firms from liquidity-risk shocks to relationship lenders when outside refinancing is impaired by asymmetric information.